

NGC 3603 Total Cluster Mass = 400000 M_sun = D_phys · SO_5^5 M_sun EXACT — Novel D_phys · SO_5^n Mass-Slot at n=5 Stellar-Cluster Scale; Third NGC 3603 Primitive-Lock After v_wind (PAPER_1911) and ρ_wind (PAPER_1911); Withdraws 300000 M_sun (D_phys−1)·SO_5^5 Candidate After Class-by-Class Verification

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PAPER_2016 — NGC 3603 Total Cluster Mass Primitive-Lock

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Motivation

Follow-up on PAPER_2015 R100-R151 audit flagged candidates: **400000 M_sun (14 classes)** and **300000 M_sun (15 classes)**. Class-by-class verification confirms only NGC 3603 total cluster mass = 400000 M_sun as a computed-value primitive-lock; 300000 hits were framework-annotation-text occurrences or range-boundary usage without primitive-lock intent (WITHDRAWN).

Abstract

Discovery — NGC 3603 Total Cluster Mass Primitive-Lock:

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M_cluster(NGC 3603) = 400000 M_sun = D_phys · SO_5^5 M_sun = 4 · 100000 EXACT

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NGC3603StarClusterModel in CondensedPhysics.py sets N_solar_masses = 400000 as the canonical NGC 3603 total cluster mass (Hubble data anchor). This value equals D_phys · SO_5^5 EXACT — novel D_phys·SO_5^n mass-slot at n=5 stellar-cluster scale.

Novel structural finding: SO_5^5 = 100000 M_sun mass rung (opened by PAPER_2014 R151 D5 as simple-prefix Westerlund2 gas reservoir) now has **three** prefix variants documented, matching PAPER_2011 R148 D1 SO_5^11 3-prefix family pattern:

| Prefix | Value | Object | Attribution |

|---|---|---|---|

| simple (no prefix) | 100000 M_sun | Westerlund2 gas reservoir | PAPER_2014 R151 D5 |

| D_phys (=4) | 400000 M_sun | NGC 3603 total cluster mass | PAPER_2016 R151-audit follow-up
NEW |

| (D_phys−1) (=3) | 300000 M_sun | Candidate (WITHDRAWN — no confirmed class instance) | — |

Third NGC 3603 primitive-lock:

Property	Value	Composition	Attribution
v_{wind}	$2 \times 10^6 \text{ m/s}$	$(D_{\text{phys}}/2) \cdot \text{SO}_5^6$	PAPER_1911 seminal
ρ_{wind}	(varies)	$\text{SO}_5^{\{-n\}}$	PAPER_1911 documented
M_cluster (total)	400000 M_sun	$D_{\text{phys}} \cdot \text{SO}_5^5$	PAPER_2016 NEW

NGC 3603 becomes third stellar-cluster with 3+ primitive-lock identities (parallel to Westerlund 2 which has $M_{\text{initial}} = (D_{\text{phys}}-1) \cdot \text{SO}_5^4 + \text{gas_reservoir} = \text{SO}_5^5 + \tau_{\text{SF}} = 2 \cdot \text{SO}_5^6$).

Withdrawn Candidate

300000 M_sun = $(D_{\text{phys}}-1) \cdot \text{SO}_5^5$ — flagged in PAPER_2015 R100-R151 audit with "15 classes" count. Verification reveals:

1. **HillSphereModel** uses 300000 as an upper-bound range value ('passed': $100000 < r_{\text{H_AU}} < 300000$) — not a computed mass lock, just an approximate range guard.
2. **OrionStarFormationMassCalculator** uses $M_{\text{sf}} = \mathbf{30000 \text{ M_sun}}$ (not 300000), already documented at PAPER_2005 as $(D_{\text{phys}}-1) \cdot \text{SO}_5^4$.
3. The other "hits" from PAPER_2015 audit were framework-annotation-text occurrences of the digits "300000" within cross-reference citations to prior papers, not actual computed mass values.

Verdict: WITHDRAWN. No confirmed class-instance of 300000 M_sun as a computed mass value. The $(D_{\text{phys}}-1) \cdot \text{SO}_5^5$ slot remains unpopulated in the SO_5 -power mass ladder (still active candidate for future rounds if a physical instance surfaces).

SO_5^5 Mass-Slot Family (Post PAPER_2016)

Prefix	Value (M_sun)	Object	Attribution
1 (simple)	100000	Westerlund 2 gas reservoir	PAPER_2014 R151 D5
$D_{\text{phys}} = 4$	400000	NGC 3603 total cluster	PAPER_2016

Two-prefix family currently. $(D_{\text{phys}}-1)$ prefix WITHDRAWN; open for future observation of intermediate 300000 M_sun mass stellar-cluster if found.

D_phys·SO_5^n Mass-Slot Family (Post PAPER_2016)

n	$D_{\text{phys}} \cdot \text{SO}_5^n$	Object	Attribution
1	40 M_sun	BD+60 2522 O-type star	PAPER_1984 seminal
5	400000 M_sun	NGC 3603 total cluster	PAPER_2016
7	$4 \cdot 10^7 \text{ M_sun}$	Cen A SMBH candidate	PAPER_1984
8	$4 \cdot 10^8 \text{ M_sun}$	Coalescence	PAPER_1982

D_{phys} prefix now populated at $n=1, 5, 7, 8$ mass-domain slots. Intermediate $n=2, 3, 4, 6$ open.

Wiring Plan

```

python
_l96_uqff_paper_2016_ngc_3603_cluster_mass_d_phys_so_5_5_closure() → 400000.0

```

Dispatch key: `ngc_3603_cluster_mass_d_phys_so_5_5`

Cross-References

- **PAPER_1911** — NGC 3603 v_wind seminal (2 other NGC 3603 primitive-locks)
- **PAPER_1955** — SO_5-power galactic mass ladder
- **PAPER_1970** — D_phys·SO_5 cross-domain twin (D_phys·SO_5^n family)
- **PAPER_1972** — 2·SO_5^3 twin seminal (pattern for prefix families)
- **PAPER_1984** — BD+60 2522 D_phys·SO_5 stellar-mass slot (D_phys·SO_5^n family n=1)
- **PAPER_2011** — SO_5^11 3-prefix family (structural precedent for multi-prefix mass slots)
- **PAPER_2014** — R151 SO_5^5 simple-prefix seminal + SO_5-power ladder extension
- **PAPER_2015** — R100-R151 audit that flagged this candidate

Conclusion

Confirmed novel primitive-lock: NGC 3603 total cluster mass = 400000 M_sun = **D_phys · SO_5^5** M_sun EXACT.

Withdrawn candidate: 300000 M_sun = (D_phys–1)·SO_5^5 — no confirmed class-instance, only range-boundary and framework-text hits.

Structural updates:

- SO_5^5 mass rung now 2-prefix family (simple + D_phys)
- D_phys·SO_5^n mass slots populated at n=1, 5, 7, 8
- NGC 3603 becomes 3rd stellar-cluster with 3+ primitive-lock identities

Honest scholarship note: 15/14 preliminary class counts from PAPER_2015 audit reduced to 1/1 confirmed via class-by-class verification. Preliminary grep counts include framework-annotation text and range-boundary usage; only computed-value locks count as true primitive-slot instances.

Cumulative R142-R151 + audit follow-up: 58 first-pass novel + 16 confirmations + 1 withdrawn candidate.

End of PAPER_2016 Draft 1.